

Logistic Regression for Multiclass classification problem

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In [1]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
%matplotlib inline
```

```
In [2]: #creating dataset
from sklearn.datasets import make_classification
X,y=make_classification(n_samples=1000, n_features=10,n_informative=3,n_c
```

```
In [3]: from sklearn.model_selection import train_test_split
X_train,X_test,y_train,y_test=train_test_split(X,y,test_size=0.30,random_
```

```
In [12]: from sklearn.linear_model import LogisticRegression
logistic=LogisticRegression()
logistic.fit(X_train,y_train)
y_pred=logistic.predict(X_test)
print(y_pred)
```

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```
In [14]: from sklearn.metrics import accuracy_score,confusion_matrix,classification_report
score=accuracy_score(y_pred,y_test)
print(score)
print(classification_report(y_pred,y_test))
print(confusion_matrix(y_pred,y_test))
```

```
0.7833333333333333
```

	precision	recall	f1-score	support
0	0.83	0.87	0.85	98
1	0.75	0.78	0.76	97
2	0.77	0.70	0.74	105
accuracy			0.78	300
macro avg	0.78	0.79	0.78	300
weighted avg	0.78	0.78	0.78	300

```
[[85  3 10]
 [ 9 76 12]
 [ 8 23 74]]
```

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In [ ]:
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